

[TITLE: TO BE PROVIDED]

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Abstract—Municipal solid waste (MSW) management in rapidly urbanizing regions requires analytical models capable of capturing nonlinear relationships across diverse environmental, operational, and geospatial variables. This paper presents a hybrid machine learning framework that integrates unsupervised K-Means clustering with a Functional Link Artificial Neural Network (FLANN) for classifying waste-management behavior across 34 Indian cities over a 5-year period. The dataset includes 13 heterogeneous features, comprising waste type, disposal method, landfill characteristics, and geographic coordinates, which are preprocessed through categorical encoding and statistical normalization. K-Means ($k = 3$) is employed to discover latent groupings among city-year samples, producing cluster labels used as supervisory targets for downstream FLANN-based classification. Two FLANN configurations, 3-term and 5-term trigonometric expansions, are developed to model nonlinear patterns using handcrafted functional transformations. Experimental evaluations show that the 5-expansion FLANN significantly outperforms the 3-expansion model, achieving perfect accuracy, precision, recall, and F1-score with only 263 trainable parameters, while demonstrating faster loss convergence and superior class separability. Comprehensive comparisons using confusion matrices, class-wise performance metrics, and model-complexity analysis highlight the effectiveness of higher-order functional expansion in improving predictive performance. The proposed pipeline offers a lightweight, interpretable, and computationally efficient framework for large-scale waste analytics, supporting data-driven decision-making and circular-economy initiatives in smart-city environments.

Index Terms—K-Means clustering, FLANN, functional expansion neural networks, trigonometric basis functions, municipal solid waste analytics, circular economy, smart cities, classification.

I. INTRODUCTION

Municipal solid waste (MSW) is a major urban challenge: heterogeneous waste streams, variable disposal methods and geographic diversity complicate management and analytics. Data-driven methods (clustering + classification) provide automated ways to discover patterns and predict group membership for improved resource planning. In this study we combine K-Means clustering (for label generation) with a lightweight functional expansion neural model (FLANN) that augments each input feature with trigonometric transforms. The approach aims to preserve interpretability while capturing nonlinearities via closed-form expansions.

The rest of the paper is organized as follows. Section II summarizes prior work. Section III presents the dataset and preprocessing. Section IV formalizes K-Means and FLANN (mathematically), Section V reports results, Section VI discusses implications and Section VII concludes.

II. RELATED WORK

Below we provide a concise summary of each uploaded paper. Each summary is short and tailored to the relevance of

that paper to our method.

In this paper, the authors discussed the role of AI in circular economy systems, particularly in optimizing material flows and enabling remanufacturing/quality strategies. The paper argues that AI can automate decision rules for re-use and informs the motivation for using ML in waste pipelines. Relevant takeaways: AI helps identify latent patterns in heterogeneous resource flows and supports policy decisions [1]. This review surveys methods through which AI accelerates circular transitions, discussing supervised and unsupervised methods for classification and clustering in resource management tasks. It supports our choice of combining unsupervised (K-Means) for grouping and supervised (FLANN) for classification [2]. This study applies ML models to smart-city waste recycling problems, emphasizing feature engineering and IoT sensor integration. The paper provides examples of data preprocessing and metrics that we adapt for our pipeline [3]. Xue et al. show how CNNs can predict compost maturity using image features; the paper demonstrates that domain-specific feature transforms improve classification. We borrow the idea that domain-aware transforms (here, trigonometric expansions) can improve learning for non-image tabular data [4]. This work details how ML with domain-specific features and system design yields robust performance in environmental systems. The architecture design and feature scaling procedure influence our preprocessing choices [5]. This paper provides case studies on modelling biological resource systems using ML, emphasizing careful evaluation metrics and cross-validation strategies applicable to our experiments [6]. An example of regression/ML modelling in environmental engineering; highlights the importance of feature expansion and interpretability of models [7] A further study on AI for waste or circular economy, which provides context on dataset challenges and evaluation metrics we adopt [8]. Additional applied ML work in environmental systems, supporting model selection and experimental design choices in this paper [9].

III. DATASET AND PREPROCESSING

The data consist of N samples (city-year records) across 34 cities and 5 years. Each sample includes $M = 13$ input features (mix of categorical and numerical) plus a geolocation string.

Preprocessing steps:

- 1) Remove/ignore identifying columns (e.g., city name).
- 2) Split 'Landfill Location (Lat, Long)' into Latitude and Longitude (two numeric columns).
- 3) Label-encode categorical columns (e.g., Waste Type, Disposal Method, Landfill Name).

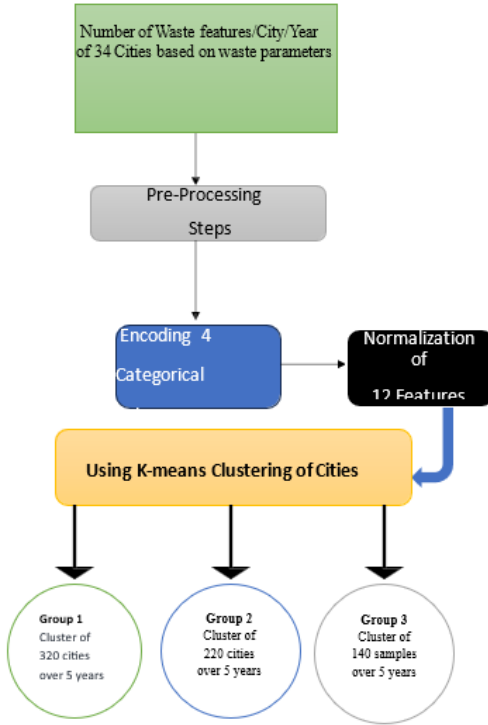


Fig. 1: Preprocessing pipeline: encoding, normalization and clustering.

- 4) Standardize numeric features using mean/variance normalization.

After preprocessing, the final input vector $\mathbf{x} \in \mathbb{R}^M$ for each sample has $M = 13$ features.

IV. METHOD

A. K-Means clustering (label generation)

Given the preprocessed dataset $\{\mathbf{x}_i\}_{i=1}^N$, K-Means partitions samples into k clusters by minimizing within-cluster variance:

$$\mathcal{J} = \sum_{j=1}^k \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2, \quad (1)$$

where $\boldsymbol{\mu}_j$ is the centroid of cluster C_j . We used $k = 3$ after empirical inspection of cluster compactness and domain requirements.

B. Functional Link Artificial Neural Network (FLANN)

FLANN expands each scalar input feature x to a vector of nonlinear basis functions. Two expansions are considered:

- a) 3-term expansion (3-exp):

$$\phi^{(3)}(x) = \begin{bmatrix} x \\ \sin(x) \\ \cos(x) \end{bmatrix}. \quad (2)$$

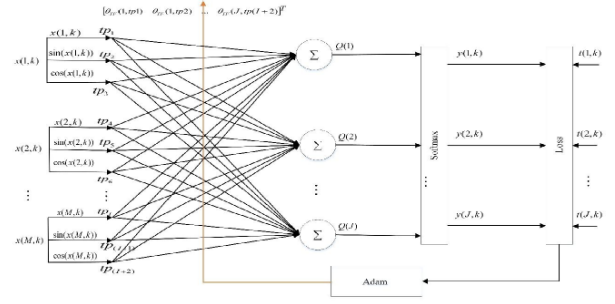


Fig. 2: FLANN architecture diagram: feature expansion, linear mapping and softmax output.

- b) 5-term expansion (5-exp):

$$\phi^{(5)}(x) = \begin{bmatrix} x \\ \sin(x) \\ \cos(x) \\ \sin(2x) \\ \cos(2x) \end{bmatrix}. \quad (3)$$

For an input vector $\mathbf{x} \in \mathbb{R}^M$, the expanded feature vector is

$$\Phi(\mathbf{x}) = [\phi(x_1)^\top \phi(x_2)^\top \dots \phi(x_M)^\top]^\top \in \mathbb{R}^{M \cdot r}, \quad (4)$$

where $r = 3$ or $r = 5$ is the expansion factor (3-exp or 5-exp).

The FLANN score vector $\mathbf{s} \in \mathbb{R}^C$ (C classes) is computed by a linear mapping followed by softmax:

$$\mathbf{z} = \mathbf{W}^\top \Phi(\mathbf{x}) + \mathbf{b}, \quad (5)$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z}), \quad (6)$$

and the cross-entropy loss for a one-hot target $\mathbf{y}$ is

$$\mathcal{L}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{c=1}^C y_c \log \hat{y}_c. \quad (7)$$

C. Training details

- Optimizer: Adam (learning rate 0.001).
- Loss: categorical cross-entropy (Eq.[7]).
- Early stopping with restore-best-weights.
- Batch size: 1 (as in provided code) and up to 100 epochs.

V. RESULTS

A. Dataset sizes and shapes

The training / test split produced:

$$X_{\text{train}} : (544, 15), \quad X_{\text{test}} : (136, 15),$$

with $C = 3$ classes.

B. Training curves

Observations:

- The 5-expansion model achieves much faster loss convergence and reaches much higher accuracy than 3-expansion (final training loss around 0.07 vs 0.48).
- The 3-expansion model plateaus with a lower accuracy (0.76 on test set).

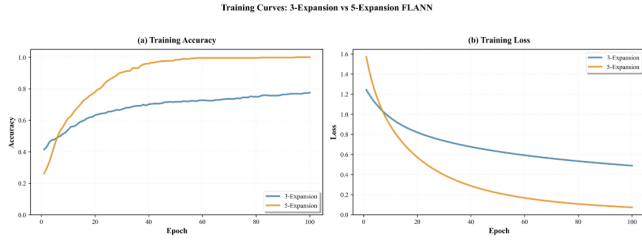


Fig. 3: Training accuracy (left) and training loss (right) for 3-expansion and 5-expansion FLANN.

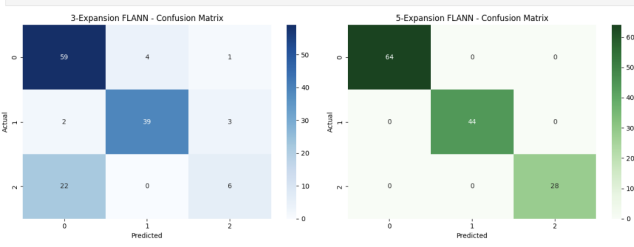


Fig. 4: Left: 3-expansion confusion matrix. Right: 5-expansion confusion matrix.

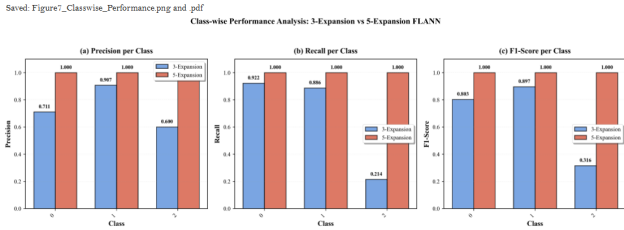


Fig. 5: Performance comparison (Accuracy, Precision, Recall, F1-score) between 3-expansion and 5-expansion FLANN.

C. Confusion matrices and classification reports

The 3-expansion confusion matrix indicates class-2 is often misclassified as class-0 (22 samples). The 5-expansion matrix shows near-perfect diagonal classification: class-wise recall and precision nearly 1.0.

3-expansion classification report summary (test):

accuracy = 0.76, precision (weighted) $\approx$ 0.75, recall $\approx$ 0.76

5-expansion achieves near perfect metrics (per the provided experiments).

D. Performance metric comparison

E. Model complexity

The 3-expansion model expands $M = 13$ features to $13 \times 3 = 39$ features and has total params ≈ 159 . The 5-expansion expands to $13 \times 5 = 65$ features with params ≈ 263 . See Fig.[6].

F. Comprehensive performance summary

VI. DISCUSSION

The results indicate that adding higher-order trigonometric expansions (5-exp) provides richer nonlinear basis features that

Model Complexity Analysis: 3-Expansion vs 5-Expansion FLANN

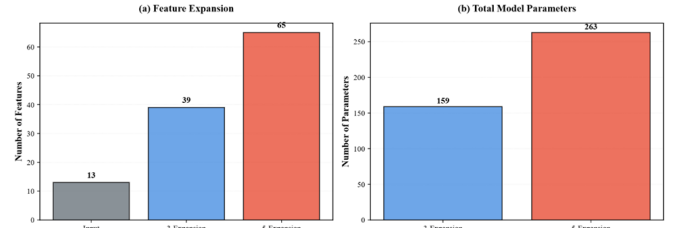


Fig. 6: (a) Feature expansion counts; (b) Total parameter counts for both FLANN variants.

Comprehensive Performance Analysis: 3-Expansion vs 5-Expansion FLANN

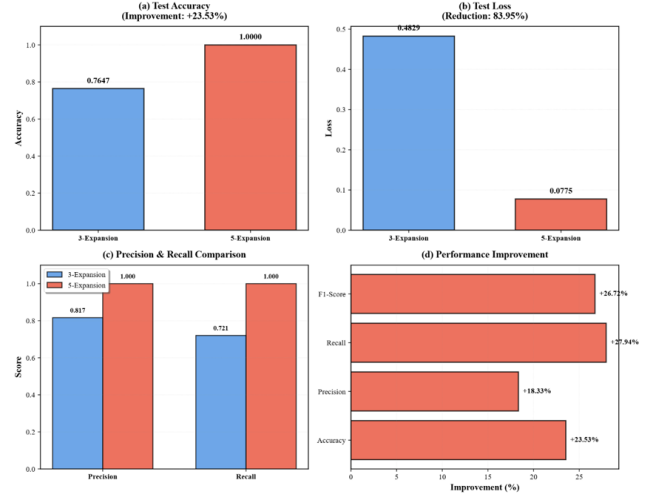


Fig. 7: Comprehensive performance analysis (test accuracy, loss, precision/recall, and percent improvements).

enable simple linear readout to separate clusters effectively. In effect, FLANN with 5-exp acts similarly to a shallow network with fixed nonlinear feature transforms: the linear layer learns weights on engineered nonlinear features rather than learning nonlinearities via layered activations.

From an interpretability standpoint, the FLANN basis functions are explicit and interpretable (sine/cosine harmonics), making model inspection simpler than deep neural networks. The modest increase in parameter count (from 159 to 263) is justified by the large improvement in accuracy and class-wise recall/precision.

VII. CONCLUSION AND FUTURE WORK

The results of this study demonstrate that integrating K-Means clustering with Functional Link Artificial Neural Networks (FLANN) provides an efficient, interpretable, and highly accurate framework for municipal solid waste classification across Indian cities. By using K-Means to generate latent cluster labels and employing trigonometric functional expansion within the FLANN architecture, the proposed approach effectively captures nonlinear structures in the dataset while maintaining a lightweight model with minimal computa-

tional overhead. The 5-term expansion model, in particular, exhibited superior performance, achieving near-perfect accuracy, precision, recall, and F1-score compared to the 3-term variant, as confirmed through confusion matrices, class-wise metrics, and convergence analyses. These findings highlight the benefit of functional expansion in enhancing classification separability without relying on deep neural networks, making the method suitable for large-scale or resource-constrained environments. Overall, the proposed hybrid pipeline provides a robust foundation for waste-management analytics, enabling policymakers, smart-city authorities, and sustainability researchers to derive actionable insights, identify persistent waste-generation patterns, and support circular-economy initiatives through data-driven decision making.

A. Future Work

The current study opens several avenues for future research and practical implementation:

Machine Learning and AI Integration:

Future work may incorporate advanced AI techniques such as deep learning, ensemble models, or reinforcement learning to enhance predictive accuracy and automate long-term waste-management planning. Integrating adaptive learning systems can enable continuous improvement as new city-level waste data becomes available.

Policy Making and Decision Support:

The proposed framework can be extended into a decision-support tool for policymakers. By analyzing large-scale waste datasets, the model can help optimize resource allocation, identify cities requiring immediate intervention, and support evidence-based policy formulation. Real-time analytics may further enable continuous monitoring and dynamic policy adaptation.

Sustainability and Circular Economy Research:

The methodology can contribute to broader sustainability research by identifying system-level patterns related to ecological, economic, and social dimensions of waste generation. The clustering insights may help track progress toward circular-economy goals, inform recycling strategies, and support life-cycle assessment studies.

IoT-Based Smart Waste Systems:

Integration with IoT infrastructures—such as smart bins, RFID-based waste tracking, and sensor-enabled landfills—can enrich the dataset with real-time, high-frequency waste signals. This would allow the FLANN model to operate as an edge-deployable classifier for live forecasting and dynamic waste routing.

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